

Multiple Choice (10 marks)

1. Which of the following is not a transport layer vulnerability?
 - (a) Mishandling of undefined, poorly defined variables
 - (b) The Vulnerability that allows "fingerprinting" & other enumeration of host information
 - (c) Overloading of transport-layer mechanisms
 - (d) Unauthorized network access
2. Encryption and decryption provide secrecy, or confidentiality, but not
 - (a) Authentication
 - (b) Integrity
 - (c) Privacy
 - (d) All of the above
3. A/an _____ is a program that steals your logins & passwords for instant messaging applications.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) Trojan-Downloader
 - (d) Ransom Trojan
4. Which of the following styles of fuzzer is more likely to explore paths covering every line of code in the following program?
 - (a) Generational
 - (b) Blackbox
 - (c) Whitebox
 - (d) Mutation-based
5. The AH Protocol provides source authentication and data integrity, but not
 - (a) Integrity
 - (b) Privacy
 - (c) Nonrepudiation
 - (d) Both A & C

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key between two parties is used only once to ensure security and prevent key reuse vulnerabilities.
7. True or False: WPA primarily relies on AES encryption for securing wireless communications.
8. True or False: Message authentication is a service that solely focuses on maintaining message confidentiality during transmission.
9. True or False: CBF is not a recognized block cipher operating mode in cryptographic practices.
10. True or False: Buffer-overflow vulnerabilities may persist in applications if boundary checks are not implemented effectively.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the Tor browser in providing anonymous access to the Tor network, and analyze its implications for user privacy and cybersecurity.
12. Discuss the role of address sanitizer (ASAN) in memory error detection during fuzzing. How does it enhance the identification of memory errors in a program?

End of Paper